- Knowledge gaps which observations should fill:

- Grid resolution in models

- Fast and small scale processes not resolved

- Microphysics (clouds)

- Even if we did know all the physics, it would be far too expensive to simulate

- Turbulence and convection (also important for clouds)

Appendix: high-resolution models are great, but...

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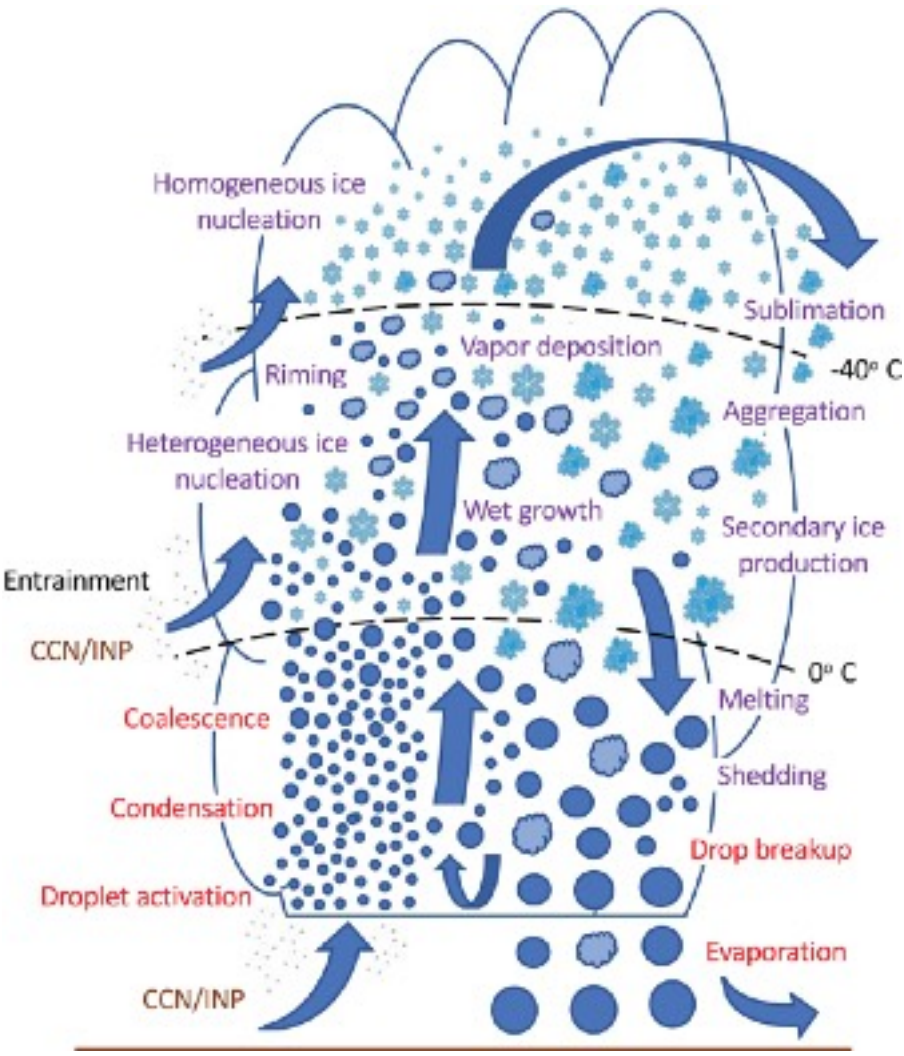
Fig.1: Morrison, H., et al. (2020). *Confronting the Challenge of Modeling Cloud and Precipitation Microphysics.* https://doi.org/10.1029/2019MS001689


Homogeneous ice nucleation
Sublimation
Vapor deposition
Riming
-40° C
Aggregation
Heterogeneous ice nucleation
Wet growth
Secondary ice production
Entrainment
CCN/INP
0° C
Melting
Coalescence
Shedding
Condensation
Drop breakup
Droplet activation
Evaporation
CCN/INP

Appendix: high-resolution models are great, but…

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  - Grid resolution in models
    - Fast and small scale processes not resolved
  - Microphysics (clouds)
    - Even if we did know all the physics, it would be far too expensive to simulate
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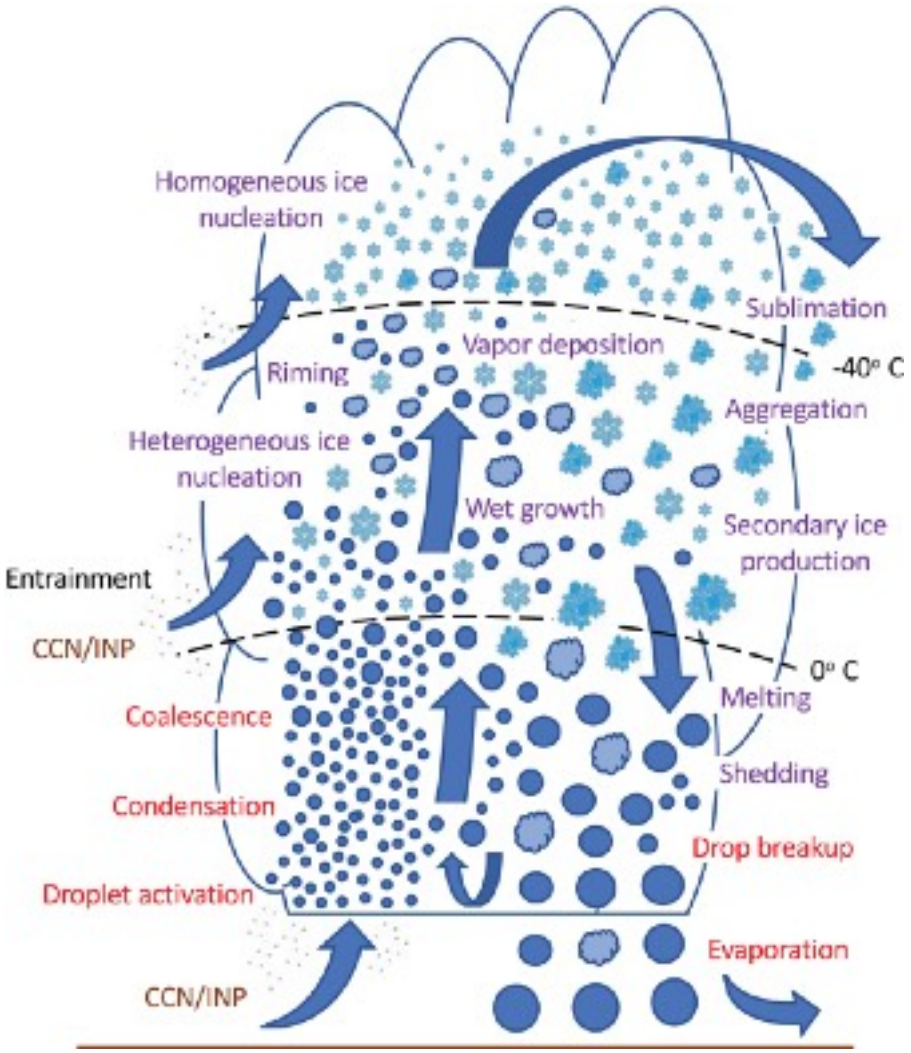

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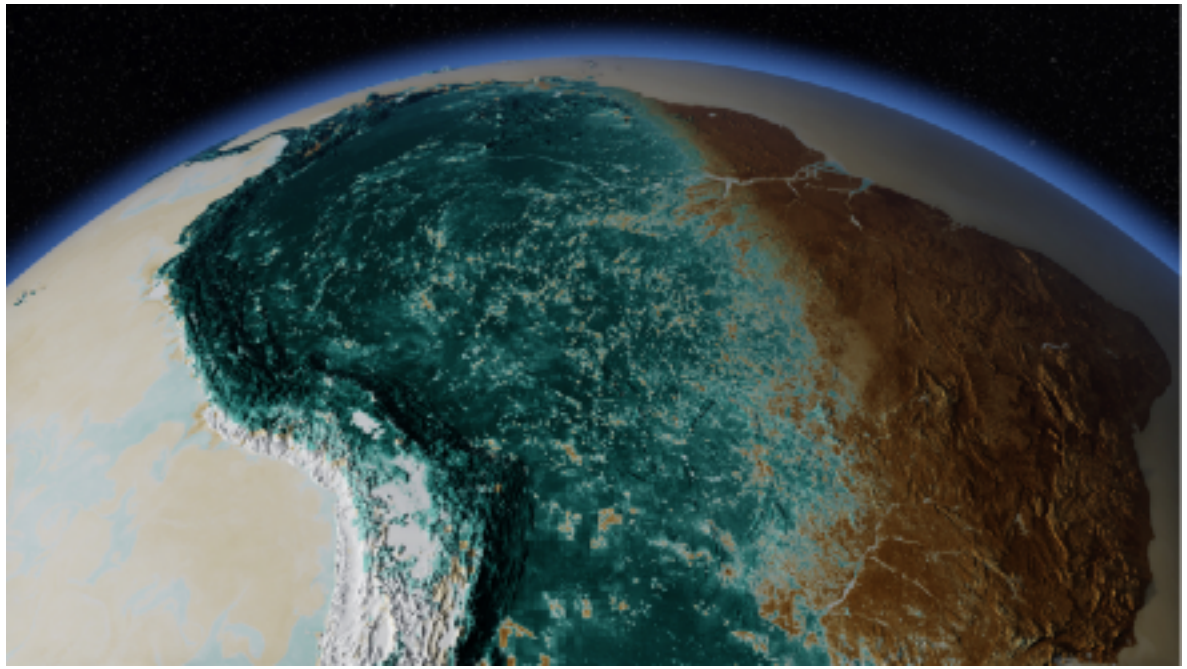
Appendix: we need models:



CO_2 flux over the Amazon basin from 1.25-km simulation